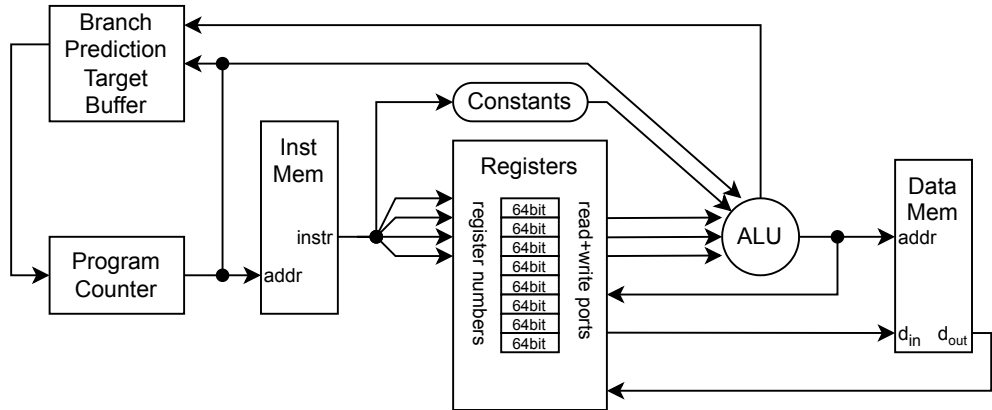
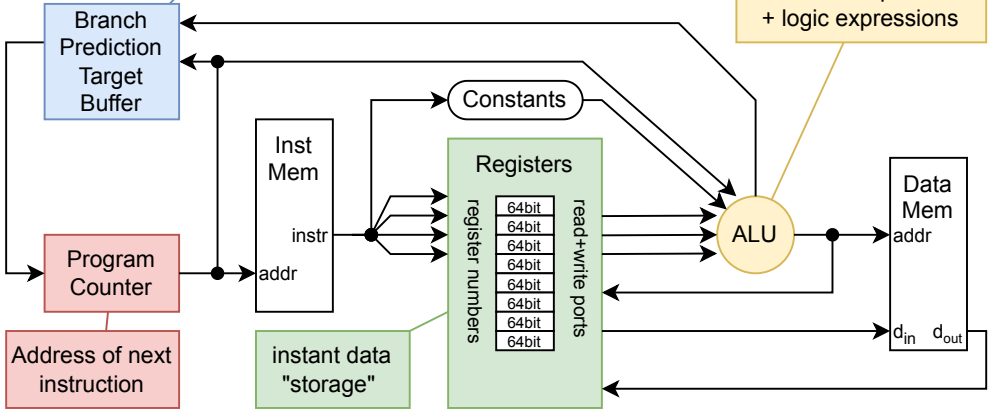




Determines which address is next

Does all computations
+ logic expressions



Program Counter

Address of next instruction

Branch Prediction Target Buffer

Instr Mem

instr

addr

instant data "storage"

Registers

register numbers

64bit
64bit
64bit
64bit
64bit
64bit
64bit
64bit

read+write ports

Constants

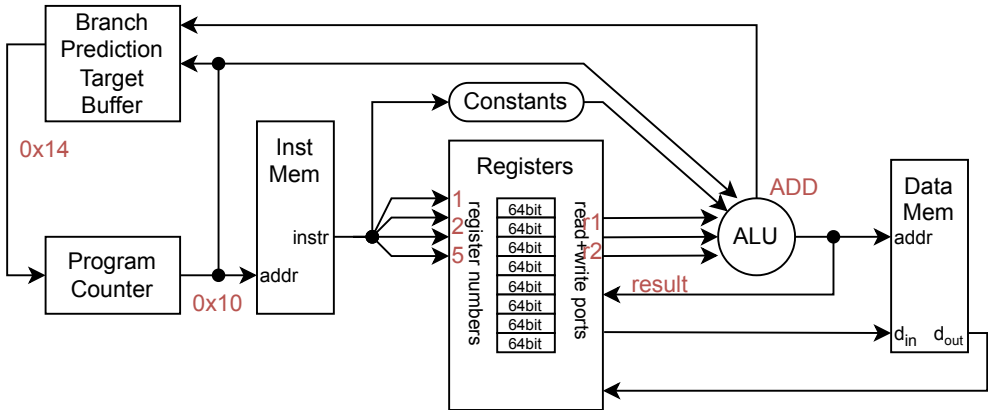
ALU

Data Mem

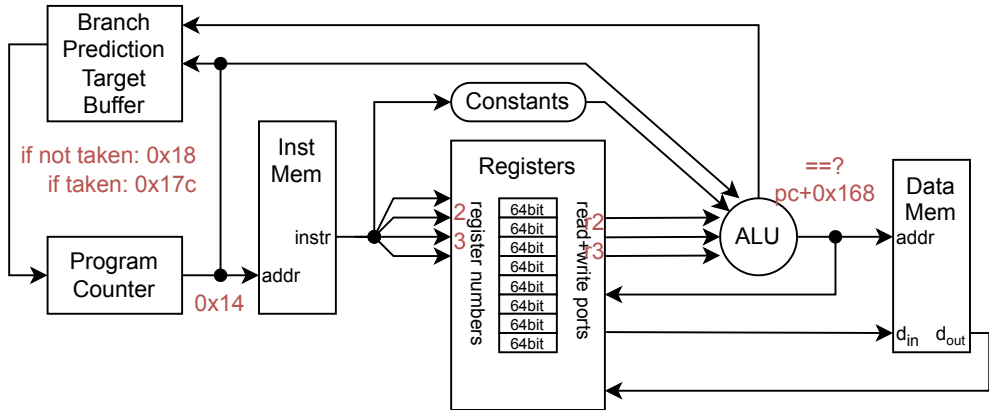
addr

d_{in} d_{out}

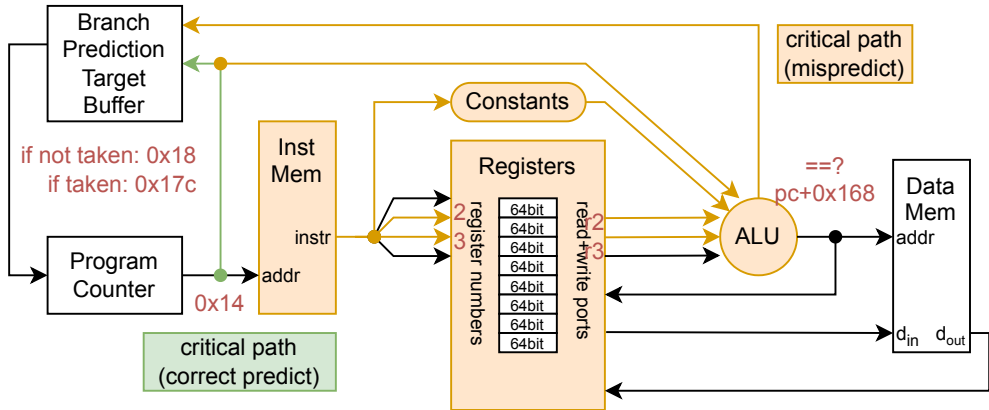
Inst @0x10: ADD r5 = r1 + r2, 4 byte long



Inst @0x14: BEQ r2 == r3, +0x168, 4 byte long



Inst @0x14: BEQ r2 == r3, +0x168, 4 byte long

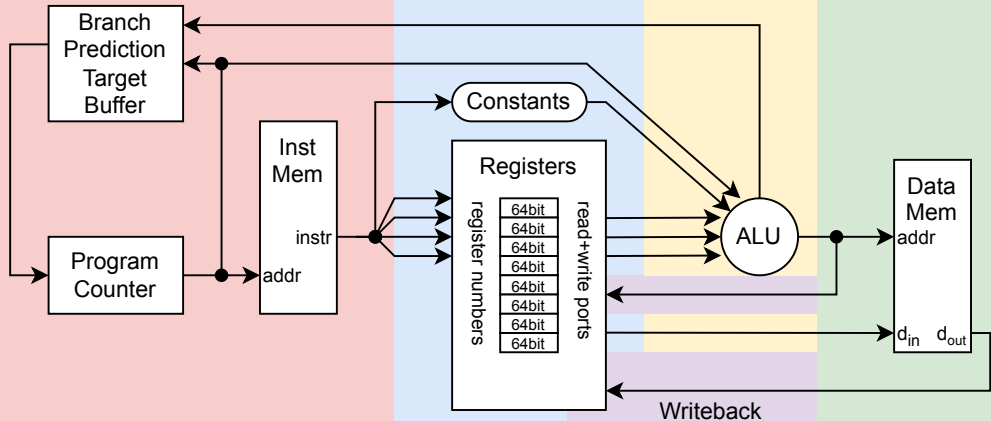


Instruction Fetch

Instruction Decode

Execute

Memory



Inst @0x18: CMOV r4, r3, r6: "overwrite if"

